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### 6.3.1 Initial insights

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put a  
superscript on it to  
indicate where it  
goes came from. And  
since we're going to  
extend this to vectors  
with more than two  
elements, we'll  
actually end up with  
more  
of these epsilons as  
well, so we're going  
to put a superscript  
on here as  
well. And notice that  
was that all these  
epsilon in magnitude  
are bounded by the  
machine  
epsilon. Okay? Now,  
at this point you can  
distribute this error  
term so that we

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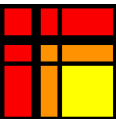
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Reading assignment

1/1 point (graded)



Read Unit 6.3.1 of the notes. [\[LINK\]](#)

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Homework 6.3.3.1

1/1 point (graded)  
Consider

$$x = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} \text{ and } x = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix}$$

and the computation

$$\kappa := x^T y$$

computed in the order indicated by

$$\kappa := (\chi_0 \psi_0 + \chi_1 \psi_1) + \chi_2 \psi_2.$$

Employ the SCM to derive a result similar to that given in Example 6.3.1.1 for the computed result  $\check{\kappa}$ .

Which of the following holds: (where  $|\epsilon_*^{(0)}|, |\epsilon_*^{(1)}|, |\epsilon_+^{(1)}|, |\epsilon_*^{(2)}|, |\epsilon_+^{(2)}| \leq \epsilon_{\max}$ )  
Mark all correct answers

- ☒  $\check{\kappa} = [(\chi_0 \psi_0 + \chi_1 \psi_1) + \chi_2 \psi_2]$
- ☒  $\check{\kappa} = [[\chi_0 \psi_0 + \chi_1 \psi_1] + [\chi_2 \psi_2]]$
- ☒  $\check{\kappa} = \left[ \left[ \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \right] + [\chi_2 \psi_2] \right]$
- ☒  $\check{\kappa} = \left( \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)}) (1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)}) (1 + \epsilon_+^{(1)}) \\ + \chi_2 \psi_2 (1 + \epsilon_*^{(2)}) \end{pmatrix} (1 + \epsilon_+^{(2)}) \right)$
- ☒  $\check{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)}) (1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)}) (1 + \epsilon_+^{(1)}) \\ + \chi_2 \psi_2 (1 + \epsilon_*^{(2)}) \end{pmatrix} (1 + \epsilon_+^{(2)})$
- ☒  $\check{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)}) (1 + \epsilon_+^{(1)}) (1 + \epsilon_+^{(2)}) & & \\ & 0 & (1 + \epsilon_*^{(1)}) \\ & 0 & \end{pmatrix}$



Solution:

For a complete solution, see [the notes](#).

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